DTMC 第一次作业答案

Problem 1

由条件知蚂蚁在$n+1$时刻的位置X_{n+1}只与n时刻的位置X_n有关系，而与X_{n-1}，X_{n-2}，…，X_1，X_0独立，即：

$$P\left(X_{n+1}=j\middle|X_n=i,X_{n-1}=i_1,...,X_1=i_{n-1},X_0=i_n\right)=P\left(X_{n+1}=j\middle|X_n=i\right)=p_{ij}.$$

所以$\{X_n,n\geq 0\}$为 DTMC，其 TPM 为：

$$P=\begin{matrix}1\\2\\3\\4\\5\\6\end{matrix}\begin{bmatrix}0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0\\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0\\ \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} & 0\\ 0 & 0 & 1 & 0 & 0 & 0\\ 0 & 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2}\\ 0 & 0 & 0 & 0 & 1 & 0\end{bmatrix}$$

Problem 2

（1）

因为$X_{n+1}=\begin{cases}(X_n-Y_{n+1})^+,\ s\leq X_n\leq S\\ (S-Y_{n+1})^+,\ 0\leq X_n<s\end{cases}$，其中$w^+=\begin{cases}w,\ w\geq 0\\ 0,\ w<0\end{cases}$，且$Y_{n+1}$与$X_1$，$X_2$,…，$X_n$相互独立。故$X_{n+1}$只依赖于现状$X_n$，独立于过去的状态$X_{n-1},...,X_1$，即

$X_{n+1}|(X_n,X_{n-1},...,X_1)\overset{d}{=}X_{n+1}|X_n$(满足 Markov 性)，所以$\{X_n,n\geq 1\}$为 DTMC.

（2）

当$j\geq 1$时，

$$p_{ij}=P\left(X_{n+1}=j\middle|X_n=i\right)$$

$$=\begin{cases}P\left(\left[X_n-Y_{n+1}\right]^+=j\middle|X_n=i\right)=P\left(Y_{n+1}=i-j\right)=a_{i-j},s\leq i\leq S,j\leq i\\ P\left(\left[S-Y_{n+1}\right]^+=j\middle|X_n=i\right)=P\left(Y_{n+1}=S-j\right)=a_{S-j},0\leq i<s,j\leq S\end{cases}$$

当$j=0$时，

$$p_{ij}=P\left(X_{n+1}=j\middle|X_n=i\right)$$

$$= \begin{cases} P\left(\left[X_n - Y_{n+1}\right]^+ = 0 \middle| X_n = i\right) = P\left(Y_{n+1} \geq i\right) = \sum_{k=i}^{+\infty} a_k, s \leq i \leq S \\ P\left(\left[S - Y_{n+1}\right]^+ = 0 \middle| X_n = i\right) = P\left(Y_{n+1} \geq S\right) = \sum_{k=S}^{+\infty} a_k, 0 \leq i < s \end{cases}.$$

所以 TPM 为

$$P = \begin{bmatrix} \sum_{k=S}^{+\infty} a_k & a_{S-1} & a_{S-2} & \cdots\cdots & a_{S-s} & \cdots & a_0 & 0 & \cdots \\ \sum_{k=S}^{+\infty} a_k & a_{S-1} & a_{S-2} & \cdots\cdots & a_{S-s} & \cdots & a_0 & 0 & \cdots \\ \cdots & \cdots & \cdots & \cdots\cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \sum_{k=S}^{+\infty} a_k & a_{S-1} & a_{S-2} & \cdots\cdots & a_{S-s} & \cdots & a_0 & 0 & \cdots \\ \sum_{k=s}^{+\infty} a_k & a_{s-1} & a_{s-2} & \cdots\cdots & a_0 & \cdots & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \cdots\cdots & \vdots & \cdots & \vdots & \vdots & \cdots \\ \sum_{k=i}^{+\infty} a_k & a_{i-1} & a_{i-2} & \cdots\cdots & a_{i-s} & \cdots & a_0 & 0 & 0 \cdots \\ \vdots & \vdots & \vdots & \cdots\cdots & \vdots & \cdots & & \vdots & \vdots \cdots \\ \sum_{k=S}^{+\infty} a_k & a_{S-1} & a_{S-2} & \cdots\cdots & a_{S-s} & \cdots & a_0 & 0 & \cdots \end{bmatrix}.$$

Problem 3

记X_{n+1}表示第$n+1$次取、放球后第 1 个坛子中的白球数，由条件知：X_{n+1}只依赖于上一次取、放球后第 1 个坛子中的白球数X_n，而与$X_{n-1}, X_{n-2}, \cdots, X_0$相互独立。故$\{X_n, n \geq 0\}$为 DTMC，状态空间 S={0,1,2,3}，其 TPM 为：

$$P = \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{1}{9} & \frac{4}{9} & \frac{4}{9} & 0 \\ 0 & \frac{4}{9} & \frac{4}{9} & \frac{1}{9} \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Problem 4

记X_n表示第n天的天气情况，由条件知X_n依赖于X_{n-1}，X_{n-2}，X_{n-3}，所以$\{X_n, n \geq 1\}$不是 DTMC.

构造三维随机变量：$Y_n = (X_n, X_{n-1}, X_{n-2})$，则随机过程$\{Y_n, n \geq 3\}$的状态空间为

$$S = \{(1,1,1), (1,1,0), (1,0,1), (1,0,0), (0,1,1), (0,1,0), (0,0,1), (0,0,0)\}$$

其中 1 表示下雨，0 表示不下雨。由条件知：Y_{n+1}只依赖于Y_n，而与$Y_{n-1}, Y_{n-2}, \cdots, Y_3$

相互独立。故$\{Y_n, n \geq 3\}$为 DTMC 其一步状态转移概率为

$$P\left(Y_n = (1,1,1) \middle| Y_{n-1} = (1,1,1)\right)$$

$$= P\left(X_n = 1, X_{n-1} = 1, X_{n-2} = 1 \middle| X_{n-1} = 1, X_{n-2} = 1, X_{n-3} = 1\right)$$

$$= P\left(X_n = 1 \middle| X_{n-1} = X_{n-2} = X_{n-3} = 1\right) = 0.8;$$

$$P\left(Y_n = (1,0,0) \middle| Y_{n-1} = (0,0,0)\right)$$

$$= P\left(X_n = 1, X_{n-1} = 0, X_{n-2} = 0 \middle| X_{n-1} = 0, X_{n-2} = 0, X_{n-3} = 0\right)$$

$$= P\left(X_n = 1 \middle| X_{n-1} = X_{n-2} = X_{n-3} = 0\right) = 0.2;$$

$$P\left(Y_n = (1,1,1) \middle| Y_{n-1} = (1,1,0)\right)$$

$$= P\left(X_n = 1, X_{n-1} = 1, X_{n-2} = 1 \middle| X_{n-1} = 1, X_{n-2} = 1, X_{n-3} = 0\right)$$

$$= P\left(X_n = 1 \middle| X_{n-1} = X_{n-2} = 1,\ X_{n-3} = 0\right) = 0.6;$$

$$P\left(Y_n = (1,1,0) \middle| Y_{n-1} = (1,0,0)\right)$$

$$= P\left(X_n = 1, X_{n-1} = 1, X_{n-2} = 0 \middle| X_{n-1} = 1, X_{n-2} = 0, X_{n-3} = 0\right)$$

$$= P\left(X_n = 1 \middle| X_{n-1} = 1, X_{n-2} = 0,\ X_{n-3} = 0\right) = 0.6;$$

$$P\left(Y_n = (1,1,0) \middle| Y_{n-1} = (1,0,1)\right)$$

$$= P\left(X_n = 1, X_{n-1} = 1, X_{n-2} = 0 \middle| X_{n-1} = 1, X_{n-2} = 0, X_{n-3} = 1\right)$$

$$= P\left(X_n = 1 \middle| X_{n-1} = 1, X_{n-2} = 0,\ X_{n-3} = 1\right) = 0.6;$$

$$P\left(Y_n = (0,0,1) \middle| Y_{n-1} = (0,1,1)\right)$$

$$= P\left(X_n = 0, X_{n-1} = 0, X_{n-2} = 1 \middle| X_{n-1} = 0, X_{n-2} = 1, X_{n-3} = 1\right)$$

$$= P\left(X_n = 0 \middle| X_{n-1} = 0, X_{n-2} = 1,\ X_{n-3} = 1\right) = 0.6;$$

$$P\left(Y_n = (0,0,1) \middle| Y_{n-1} = (0,1,0)\right)$$

$$= P\left(X_n = 0, X_{n-1} = 0, X_{n-2} = 1 \middle| X_{n-1} = 0, X_{n-2} = 1, X_{n-3} = 0\right)$$

$$= P\left(X_n = 0 \middle| X_{n-1} = 0, X_{n-2} = 1,\ X_{n-3} = 0\right) = 0.6;$$

$$P\left(Y_n=(0,0,0)\middle|Y_{n-1}=(0,0,1)\right)$$

$$=P\left(X_n=0,X_{n-1}=0,X_{n-2}=0\middle|X_{n-1}=0,X_{n-2}=0,X_{n-3}=1\right)$$

$$=P\left(X_n=0\middle|X_{n-1}=0,X_{n-2}=0,\ X_{n-3}=1\right)=0.6;$$

所以$\{Y_n,n\geq 3\}$的 TPM 为：

$$P=\begin{matrix}(1,1,1)\\(1,1,0)\\(1,0,1)\\(1,0,0)\\(0,1,1)\\(0,1,0)\\(0,0,1)\\(0,0,0)\end{matrix}\begin{bmatrix}0.8&0&0&0&0.2&0&0&0\\0.6&0&0&0&0.4&0&0&0\\0&0.6&0&0&0&0.4&0&0\\0&0.6&0&0&0&0.4&0&0\\0&0&0.4&0&0&0&0.6&0\\0&0&0.4&0&0&0&0.6&0\\0&0&0&0.4&0&0&0&0.6\\0&0&0&0.2&0&0&0&0.8\end{bmatrix}$$

Problem 5

（1）建构 DTMC 模型；

记X_n表示第n局赌博后，赌徒的财富，记$Y_i=\begin{cases}1,\text{在第}i\text{局赌博中胜}\\-1,\text{在第}i\text{局赌博中败}\end{cases}$

则$\{Y_i,i\geq 1\}$为 i.i.d. r.v.s. 且$X_n=X_0+\sum_{i=1}^{n}Y_i$, 因为$X_{n+1}=X_n+Y_{n+1}$, 其中$Y_{n+1}$独立于$X_{n-1}$，$X_{n-2}...X_1$，$X_0$，故$X_{n+1}$只依赖于$X_n$，而与$X_i,i\leq n-1$ 相互独立，

所以$\{X_n,n\geq 0\}$为 DTMC，其状态空间$S=\{0,1,2,3\}$，由条件知：其 TPM 为 $P=$

$$\begin{matrix}0\\1\\2\\3\end{matrix}\begin{bmatrix}1&0&0&0\\\frac{1}{2}&0&\frac{1}{2}&0\\0&\frac{1}{2}&0&\frac{1}{2}\\0&0&0&1\end{bmatrix}.$$

（2）利用模型解决问题；

由于$P^4=\begin{bmatrix}1&0&0&0\\\frac{5}{8}&\frac{1}{16}&0&\frac{5}{16}\\\frac{5}{16}&0&\frac{1}{16}&\frac{5}{8}\\0&0&0&1\end{bmatrix}$,

所以 P(时刻 4 赌徒的财富为 0)$=P\left(X_4=0\middle|X_0=2\right)=P_{20}^{(4)}=\frac{5}{16}$.

Problem 6

（1）建构 DTMC 模型；

记X_n表示质点第n次移动后所在的位置，由于X_{n+1}只依赖于上一次移动后的位置X_n，而与$X_i, i \le n-1$ 相互独立，所以$\{X_n, n \ge 0\}$为 DTMC，状态空间为$S = \{-2, -1, 0, 1, 2\}$, 由条件知：其 TPM 为

$$P = \begin{array}{c} \\ -2 \\ -1 \\ 0 \\ 1 \\ \\ 2 \end{array} \begin{array}{c} \begin{array}{ccccc} -2 & -1 & 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

（2）利用模型解决问题；

由于 $P^3 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0.5185 & 0.1481 & 0.1852 & 0.1111 & ※ \\ ※ & ※ & 0.2593 & ※ & ※ \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}$，$\rightarrow P_{00}^{(3)}$

所以$P_{00}^{(3)}$=0.2593.

Problem 7

$$X_{n+1} = \left(\sum_{k=1}^{n+1} Y_k\right)^2 = \left(\sum_{k=1}^{n} Y_k + Y_{n+1}\right)^2 = \left(\sum_{k=1}^{n} Y_k\right)^2 + 2Y_{n+1}\left(\sum_{k=1}^{n} Y_k\right) + Y_{n+1}^2$$
$$= X_n + 2\sqrt{X_n}Y_{n+1} + Y_{n+1}^2.$$

由条件知：Y_{k+1}独立于$X_n, X_{n-1}, ..., X_1$，故X_{n+1}只依赖于X_n，而与$X_i, i \le n-1$ 相互独立，所以$\{X_n, n \ge 1\}$为 DTMC，状态空间为$S = \{0^2, 1^2, 2^2, ...\}$.

一步转移概率为：

$$P_{ij} = P\left(X_{n+1} = j \middle| X_n = i\right) = P\left(X_n + 2\sqrt{X_n}Y_{n+1} + Y_{n+1}^2 = j \middle| X_n = i\right)$$

$$= P\left(i + 2\sqrt{i}Y_{n+1} + Y_{n+1}^2 = j \middle| X_n = i\right)$$

$$= P\left(Y_{n+1}^2 + 2\sqrt{i}Y_{n+1} + i = j\right)\left(\text{因为}X_n\text{独立于}Y_{n+1}\right) = P\left(Y_{n+1} = \sqrt{j} - \sqrt{i}\right)$$

$$= \begin{cases} a_{\sqrt{j}-\sqrt{i}}, & j \geq i \\ 0, & j < i \end{cases}.$$

Problem 8

（1）建构 DTMC 模型；

记X_n表示第n局比赛后甲的得分，由于各局比赛相互独立，所以甲在第$n+1$局比赛后的得分X_{n+1}只依赖于X_n，而与$X_i, i \leq n-1$相互独立，所以$\{X_n, n \geq 1\}$为DTMC，状态空间$S = \{-2, -1, 0, 1, 2\}$，由条件知：其 TPM 为

$$P = \begin{matrix} -2 \\ -2 \\ 0 \\ 1 \\ 2 \end{matrix}\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ q & r & p & 0 & 0 \\ 0 & q & r & p & 0 \\ 0 & 0 & q & r & p \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

（2）利用模型解决问题；

由于$P^2 = \begin{matrix} -2 \\ -2 \\ 0 \\ 1 \\ 2 \end{matrix}\begin{bmatrix} & & & & \\ & & & & \\ & & & & \\ 0 & q^2 & 2qr & pq + r^2 & p + pr \end{bmatrix}$.

故 P(2 局内结束比赛$|X_1 = 1) = P_{12}^{(2)} + P_{1-2}^{(2)} = (p + rp) + 0 = p + rp.$